

Operating Systems - Sample Study Notes

Introduction to Operating Systems

An Operating System (OS) is system software that manages computer hardware and software resources. Its primary goals are convenience, efficiency, and the ability to evolve. Examples include Windows, Linux, macOS, Android, and iOS.

Process Management

A process is a program in execution. The operating system is responsible for creating, scheduling, synchronizing, and terminating processes. Key concepts include Process Control Block (PCB), process states, and context switching.

CPU Scheduling

CPU scheduling determines which process gets CPU time. Common algorithms: - First Come First Serve (FCFS) - Shortest Job First (SJF) - Round Robin (RR) - Priority Scheduling Performance metrics include turnaround time, waiting time, and response time.

Threads

Threads are lightweight units of execution within a process. Advantages include improved responsiveness, resource sharing, and parallelism. Types include user-level threads and kernel-level threads.

Deadlocks

A deadlock occurs when two or more processes wait indefinitely for resources held by one another. Necessary conditions: 1. Mutual Exclusion 2. Hold and Wait 3. No Preemption 4. Circular Wait Strategies: - Prevention - Avoidance - Detection and Recovery

Memory Management

Memory management controls allocation and deallocation of memory. Concepts: - Paging - Segmentation - Virtual Memory - Demand Paging Virtual memory allows programs larger than physical RAM to execute.

File Systems

A file system organizes and stores data on storage devices. Responsibilities include: - File creation and deletion - Directory management - Access control Examples: NTFS, FAT32, ext4.

Key Takeaways

1. The OS acts as an intermediary between users and hardware. 2. CPU scheduling improves system utilization. 3. Threads enable efficient concurrent execution. 4. Deadlocks must be prevented or resolved. 5. Virtual memory improves memory utilization. 6. File systems provide organized persistent storage.